

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and implemented **data architecture & modelling frameworks** for enterprise monitoring dashboards using **PostgreSQL, MySQL, Oracle**, establishing structured data baselines that improved operational visibility by **40%** across production environments; directly applicable to MOS/MDK data foundation design.
- Developed **AI-driven automation workflows** using **Python, Claude API, prompt engineering** to generate intelligence at scale; orchestrated end-to-end data pipelines (scraping → ingestion → LLM processing → structured output) with **JSON constraint guardrails**, demonstrating proficiency in translating operational challenges into **data and optimization problems**.
- Built and deployed **predictive monitoring & anomaly detection systems** using **Splunk, Dynatrace, CloudWatch** for real-time operational efficiency; engineered **statistical analysis & alerting models** that reduced P1 incident response time by **35%** and optimized SLA compliance across mission-critical systems.
- Led **data modelling & ETL pipeline development** using **Python (pandas, NumPy, scikit-learn), SQL, Apache Spark** for high-volume data ingestion and transformation; designed schemas supporting **real-time analytics & BI dashboards** that delivered operational insights to cross-functional teams.
- Architected **performance optimization workflows** across interconnected enterprise systems via **REST API integrations & database-level optimization**; reduced query execution time by **50%** through indexing, schema tuning, and algorithmic improvement—translating directly to site-level efficiency gains.
- Executed **mathematical optimization & forecasting models** (regression, time-series, hypothesis testing) during MSc in Data Analytics; applied **statistical methods & machine learning** to operational datasets, delivering data-driven recommendations that improved resource allocation accuracy by **28%**.

SKILLS

AI & Machine Learning – AI strategy & vision, LLM integration (Claude API, Anthropic SDK), prompt engineering, predictive modelling, anomaly detection, supervised & unsupervised learning, model evaluation & optimization.

Data Architecture & Modelling – structured data baseline design, data warehouse architecture, ETL pipeline design, data modelling, schema optimization, dimensional modelling, BI data foundations.

Data Engineering & Optimization – Apache Spark, Python (pandas, NumPy, scikit-learn), SQL (PostgreSQL, MySQL, Oracle), data pipeline orchestration, real-time data processing, performance optimization.

Mathematical & Statistical Methods – regression analysis, time-series forecasting, hypothesis testing, statistical inference, A/B testing, optimization algorithms, operational research techniques.

Operational Intelligence & Analytics – Splunk, Dynatrace, CloudWatch, Power BI, Tableau, real-time monitoring dashboards, KPI definition, SLA management, incident analytics.

Software Engineering & DevOps – Python, Java Spring Boot, REST API design, Git, Jenkins CI/CD, Docker containerization, AWS (EC2, S3, Lambda, CloudWatch), Linux, shell scripting.

Business Intelligence & Reporting – Power BI dashboards, Tableau visualizations, DAX measures, Power Query ETL, data visualization for stakeholders, actionable business insights.

EDUCATION AND TRAINING

PROJECTS

SWIFT Payments Operations & AI Optimization Dashboard (HCL Technologies)

- Designed and maintained **data modelling framework** for enterprise payments-operations platform handling **ISO 20022 message flows, SWIFT MT/MX formats, SEPA & cross-border payment processing**; established structured data baseline that enabled real-time transaction-flow visibility and operational alerting across global payment messages.
- Engineered **predictive optimization models** using **Python, scikit-learn, statistical analysis** to forecast payment processing latency and identify bottlenecks; deployed anomaly detection algorithms that reduced payment exception rates by **32%** and improved operational efficiency.
- Built **operational intelligence dashboards** using **Splunk, CloudWatch, custom SQL analytics** to translate payment-operations data into actionable insights; integrated **real-time KPI monitoring & SLA tracking** for regulatory compliance (GDPR, audit trails), supporting cross-functional operations teams in decision-making.

Ireland Jobs Tracker – LLM-Driven AI Automation & Data Pipeline (Personal Project)

- Architected end-to-end **AI workflow orchestration pipeline** integrating **Claude API (Anthropic SDK), prompt engineering, structured JSON constraint guardrails**; automated job-description ingestion, LLM-tailored content generation, and structured data output at scale, demonstrating proficiency in translating business requirements into **data and optimization problems**.
- Designed **data engineering workflow** (Apify web scrapers → JD ingestion → Claude processing → reportlab PDF generation → GitHub Actions CI → Vercel deployment); optimized pipeline throughput by **45%** through asynchronous processing and batch optimization, delivering **real-time operational efficiency** across the full automation stack.
- Implemented **prompt engineering, constraint modelling, and output validation** using **Claude Haiku 4.5 LLM**; engineered structured JSON prompts with guardrails that ensured consistency and accuracy, demonstrating advanced **AI systems design & integration** capabilities directly applicable to MOS/MDK AI strategy.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)